

Disproof of conjecture 00000001260

earthking11

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Verdict: FALSE

We disprove conjecture 00000001260 as stated. The conjecture asserts a *universal* upper bound, $p(n) \leq n + 2$, for the factor complexity of every primitive substitution in the “purely substitutive class”. The Thue–Morse word, the fixed point of the primitive substitution $0 \mapsto 01, 1 \mapsto 10$, already violates this bound at $n = 3$, where $p(3) = 6 > 5 = 3 + 2$. The failure is not an isolated accident: the celebrated Tribonacci word named in the conjecture itself satisfies $p(n) = 2n + 1 > n + 2$ for every $n \geq 2$, so it too refutes the stated bound.

1 The conjecture

Quoted from `conjectures/00000001260.md`:

Definition: The factor complexity $p(n)$ of a primitive substitution counts distinct factors of length n . **Conjecture:** The optimal upper bound is $p(n) \leq n + 2$ within the purely substitutive class, attained by three-letter substitution families beyond the Tribonacci word; $p(n) = n + 1$ occurs only for Sturmian words.

The filed statement consists of a *definition* (factor complexity), a *universal bound* ($p(n) \leq n + 2$ over the class), an *attainment* claim (three-letter families attain it), and a *classification* claim ($p(n) = n + 1$ only for Sturmian words). The universal bound is the load-bearing part; we refute it, and thereby the conjecture. The second clause, $p(n) = n + 1$ for Sturmian words, is true and we confirm it below.

2 Definitions

Definition 1 (Substitution and fixed point). A *substitution* is a morphism σ on words over a finite alphabet. A word w is a *fixed point* of σ if $\sigma(w) = w$ and w begins with $\sigma(w_0)$; the fixed point is obtained as $\lim_{k \rightarrow \infty} \sigma^k(0)$ when $\sigma(0)$ begins with 0.

Definition 2 (Factor complexity). For an infinite word w , the *factor complexity* $p(n)$ is the number of distinct finite words of length n (its *factors*) occurring as a contiguous block in w .

Definition 3 (Primitive substitution). A substitution σ on a d -letter alphabet is *primitive* if there is $k \geq 1$ such that every entry of the incidence matrix M^k is strictly positive, where M_{ij} is the number of occurrences of letter j in $\sigma(i)$.

Definition 4 (Thue–Morse word). The Thue–Morse substitution on $\{0, 1\}$ is

$$\sigma_{\text{TM}}: \quad 0 \mapsto 01, \quad 1 \mapsto 10.$$

Its fixed point starting with 0 is the *Thue–Morse word* $\text{TM} = 0110100110010110 \dots$. Equivalently, the n -th letter of TM is the parity of the binary digit sum of n .

3 The Thue–Morse counterexample

Proposition 5 (Primitivity). σ_{TM} is primitive. Its incidence matrix is $M = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, every entry of which is positive; hence M^k is positive for every $k \geq 1$, so the defining condition holds already with $k = 1$.

Proof. Immediate from the definition: $\sigma_{\text{TM}}(0) = 01$ contains one 0 and one 1, and likewise $\sigma_{\text{TM}}(1) = 10$. Thus M has all entries $1 > 0$. $\square$

The Thue–Morse word is standardly *purely substitutive*: it is the fixed point of an actual substitution, and this substitution is primitive. It is a constant-length (of length 2) substitution, and the filed conjecture text imposes no restriction excluding constant-length substitutions and no restriction on the size of the alphabet.

Theorem 6 (The bound fails at $n = 3$). *The Thue–Morse word has at least six distinct factors of length 3; indeed*

$$p_{\text{TM}}(3) = 6 > 5 = 3 + 2.$$

Consequently the asserted universal bound $p(n) \leq n + 2$ is false within the purely substitutive class.

Proof. The six blocks

$$011, \quad 110, \quad 101, \quad 010, \quad 100, \quad 001$$

each occur in TM as a length-3 factor: they start at positions 0, 1, 2, 3, 4, 5 of the prefix 0110100110010110..., respectively. They are pairwise distinct, so $p_{\text{TM}}(3) \geq 6$. A direct enumeration over the prefix of length 2^{18} gives exactly 6 distinct length-3 factors, so $p_{\text{TM}}(3) = 6$. Since $6 > 5$, the bound $p(n) \leq n + 2$ fails at $n = 3$. $\square$

Corollary 7. *Conjecture 00000001260 is false: its universal upper bound is violated by a primitive, purely substitutive word.*

Remark 8 (Only a lower bound is needed). To refute a universal *upper* bound it suffices to exhibit, for a single word, a single n and a *lower* bound $p(n) > n + 2$. Each of the six blocks above is a genuine factor by construction (it starts at an explicit position), so the count 6 is a lower bound on $p_{\text{TM}}(3)$; no completeness argument (no proof that there are *at most* six) is required. This is exactly what the Lean formalisation records.

4 The complexity table for $n = 1, \dots, 15$

Counting distinct factors of TM over the prefix of length 2^{18} :

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$p(n)$	2	4	6	10	12	16	20	22	24	28	32	36	40	42	44
$n + 2$	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

The bound $p(n) \leq n + 2$ holds only at $n = 2$ (where $p(2) = 4 = 2 + 2$) and fails for every $n \geq 3$, the first failure being $p(3) = 6 > 5$. Numerical verification confirms $p(n) > n + 2$ for every $n \in [3, 300]$.

5 Bonus counterexamples

5.1 The Tribonacci word (the word the conjecture names)

The Tribonacci word is the fixed point of the primitive substitution

$$0 \mapsto 01, \quad 1 \mapsto 02, \quad 2 \mapsto 0.$$

It is a three-letter substitution, and it is precisely the “Tribonacci word” the conjecture refers to. Its factor complexity is known to be exactly linear,

$$p(n) = 2n + 1 \quad \text{for all } n \geq 1.$$

Hence $p(n) = 2n + 1 > n + 2$ for every $n \geq 2$. So the very word named in the conjecture violates the asserted universal bound, not merely some exotic example. Computation over a prefix of length 223317 reproduces $p(1..15) = 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31$, matching $2n + 1$.

5.2 The period-doubling word

The period-doubling word is the fixed point of

$$0 \mapsto 01, \quad 1 \mapsto 00.$$

Computing over a prefix of length 65536 gives

$$p(1), p(2), p(3), p(4), p(5), \dots = 2, 3, 5, 6, 8, \dots$$

so the bound $p(n) \leq n + 2$ first fails at $n = 5$, where $p(5) = 8 > 7$.

5.3 Sturmian words confirm the second clause

The Fibonacci word, fixed point of $0 \mapsto 01, 1 \mapsto 0$, is the archetypal Sturmian word, and it satisfies $p(n) = n + 1$ for all $n \geq 1$. Computation over a prefix of length 131072 gives $p(1..14) = 2, 3, 4, \dots, 15$, exactly $n + 1$. Thus the conjecture’s classification clause ($p(n) = n + 1$ only for Sturmian words) is consistent with our data; it is the universal bound that fails.

6 Reading caveat (honesty statement)

The refutation above targets the *literal universal reading* of the filed text: “the optimal upper bound is $p(n) \leq n + 2$ within the purely substitutive class”, with no alphabet restriction and no exclusion of constant-length substitutions.

- If “purely substitutive” were silently restricted to *three-letter* substitutions, or to *Pisot-type* substitutions, the Thue–Morse counterexample (a two-letter, constant-length substitution) would be excluded and the refutation would not apply. *No such restriction appears in the filed text.*
- If the conjecture were reinterpreted as a statement about the *minimal growth rate over the class* rather than a universal upper bound, then the Thue–Morse word would again be irrelevant. Again, *no such reinterpretation is present in the text*: the sentence plainly asserts a bound $p(n) \leq n + 2$ holding within the class.
- The sentence itself is garbled: it mixes a universal bound ($p(n) \leq n + 2$ “within the class”) with an *attainment* claim (“attained by three-letter substitution families beyond the Tribonacci word”), which are statements of different logical type. It is not a precise mathematical claim, and no repaired reading is supplied by the text.

Under the literal reading there is no ambiguity: the bound is false.

7 Reproducibility

The checks are carried out by `reproduce.py`, which uses the Python 3 standard library only. It builds the Thue–Morse prefix of length 2^{18} by iterated substitution, counts distinct length- n factors for $n = 1, \dots, 300$, verifies $p(3) = 6$ and $p(n) > n + 2$ for all $n \in [3, 300]$, and likewise computes the period-doubling, Tribonacci, and Fibonacci complexities:

```
$ python3 reproduce.py
... (tables and checks)
PASS: conjecture 00000001260 is FALSE.
```

It prints PASS and exits 0 exactly when every check holds.

8 Formalisation

The core of the disproof is formalised in the accompanying Lean 4 project under `lean4/`, built with `lake build` and audited by `lake env lean Check.lean`. It uses core Lean only (`import Std`, no `Mathlib`) and no `sorry`. The Thue–Morse word is defined by a structural recursion so that the concrete computation reduces under `decide`, and the key statements are:

```
theorem tm_p3_ge_6 : 6 ≤ fac3.eraseDups.length
theorem tm_p3_eq_6 : fac3.eraseDups.length = 6
theorem tm_refutes_bound : fac3.eraseDups.length > 3 + 2
theorem tm_prefix_agrees : (List.range 64).map tm = gen 6
theorem incidence_positive : ∀ r ∈ incidence, ∀ c ∈ r, 0 < c
theorem conjecture_00000001260_false :
  fac3.eraseDups.length = 6 ∧ ¬ (fac3.eraseDups.length ≤ 3 + 2)
```

The formalisation records exactly the lower bound $p(3) \geq 6$: `fac3` is the list of length-3 blocks starting at explicit positions $0, \dots, 61$, so each of its entries is a genuine factor, and `fac3.eraseDups.length` is therefore a lower bound for $p(3)$. The consistency check `tm_prefix_agrees` ties the closed-form definition of TM to the substitution-generated prefix, and `incidence_positive` witnesses primitivity of $0 \mapsto 01, 1 \mapsto 10$.

`Check.lean` prints `#print axioms` for every theorem and reports `propext` only; `incidence_positive` depends on no axioms at all. In particular it reports neither `sorryAx` nor `Lean.ofReduceBool`; no `Mathlib` is involved.

Files

File	Description
<code>README.md</code>	Verdict, bilingual quote, definitions, tables, caveat, status.
<code>main.tex</code>	This source; standalone article, <code>tectonic --outdir build main.tex</code> .
<code>build/main.pdf</code>	Compiled PDF.
<code>reproduce.py</code>	Stdlib-only reproduction; prints PASS/FAIL.
<code>lean4/</code>	Lean 4 project: <code>Main.lean</code> , <code>Check.lean</code> , <code>lakefile.toml</code> , <code>lean-toolchain</code> , <code>README.md</code> .